Conjectural permutation decoding of some AG codes*

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Abstract

We study the action of a finite group on the Riemann-Roch space of certain divisors on a specific hyperelliptic curve X defined over a finite field with "large" automorphism group G. If D and $E = P_1 + ... + P_n$ are G-equivariant divisors on X ($P_i \in X(F)$) then G acts on associated AG code $C = C(D, E)$ by permuting coordinates. This note discusses the permutation decoding of these AG codes. The main "results" are conjectures regarding the complexity of the permutation decoding of these hyperelliptic codes. The open source GAP error-correcting codes package GUAVA is used to compute examples.

1 Introduction

Let X be a plane curve defined over a finite field $F = GF(q)$ by means of some polynomial equation $f(x, y) = 0$, where f has coefficients in F. (Though strictly speaking we are identifying, by abuse of notation, X with the associated projective algebraic curve.) Let D be a formal integer linear combination of F-rational points in $X(F)$, $D = d_1Q_1+...d_rQ_r$. The Riemann-Roch space $L(D)$ is an F-vector space of rational functions in x, y on X whose zero/pole at Q_i is controlled by d_i, $1 \leq i \leq r$. (See Stichtenoth [Sti] or Hartshorne [H] for a precise definition.) Let $E = P_1 + ... + P_n$ be a sum

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of distinct distinct whose support is disjoint from that of D. The *AG code* $C = C_X(D, E)$ associated to X is defined by

$$C = \{(f(P_1), ..., f(P_n)) \mid f \in L(D)\}. \tag{1.1}$$

This is the image of $L(D)$ under the evaluation map

$$\begin{array}{c} \mathrm{eval}_E : L(D) \to F^n, \\ f \longmapsto (f(P_1), ..., f(P_n)). \end{array} \tag{1.2}$$

These codes were originally introduced by Goppa in the 1980s and have proven to be a rich source of efficient error-correcting linear block codes. The special case where D is a multiple of a single point in $X(F)$ (often the point at infinity, if it is unique, and then E is usually the sum of the remaining points) is called a *one-point AG code*.

We say that the divisor D above is *G-equivariant* if the action of G on $X(F)$ leaves D invariant. (Therefore, D must in fact be an integer linear combination of G-orbits of $X(F)$.) If D is G-equivariant then $L(D)$, and hence C, is a G-module.

This paper investigates one small class of such AG codes which seems to have an unusually large amount of symmetry. The hope is that this symmetry can be used to create a class of codes which has a very efficient decoding procedure.

Some of these ideas are part of joint research with Amy Ksir and Will Traves. For related research, see Joyner and Ksir [JK1]. See [JK2] and [JT] for further background.

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2 AG codes associated to $y^2 = x^p - x$

Properties of $y^2 = x^p - x$ are discussed for example in Göb [G].

2.1 Background on the curve

In general, if X is a curve defined over a field F with finite automorphism group $G = \mathrm{Aut}_F(X)$ then we call G **large** if $|G| > |X(F)|$.

Lemma 1 *If G is large then every point of $X(F)$ is ramified for the covering $X \to X/G$.*

Proof: Suppose $P \in X(F)$ is not ramified, so the stabilizer of P, G_P, is trivial. In this case, $|G \cdot P| = |G|/|G_P| = |G|$. But $G \cdot P \subset X(F)$ so $|G \cdot P| \leq |X(F)|$, a contradiction. $\square$

2.1.1 Case $F = GF(p)$

Let $p \geq 3$ be a prime, $F = GF(p)$, and let X denote the curve defined by

$$y^2 = x^p - x.$$

This has genus $\frac{p-1}{2}$ and $p+1$ F-rational points[1]. We assume that the automorphism group $G = \mathrm{Aut}_F(X)$ is a central 2-fold cover of $PSL(2,p)$, we have a short exact sequence,

$$1 \to Z \to G \to PSL_2(p) \to 1, \tag{2.1}$$

where Z denotes the center of G (Z is generated by the hyperelliptic involution). The following transformations are elements of G:

$$\begin{array}{cc} \gamma_1 = \left\{ \begin{array}{c} x \longmapsto x, \\ y \longmapsto -y, \end{array} \right. , & \gamma_2 = \gamma_2(a) = \left\{ \begin{array}{c} x \longmapsto a^2 x, \\ y \longmapsto ay, \end{array} \right. \\ \gamma_3 = \left\{ \begin{array}{c} x \longmapsto x+1, \\ y \longmapsto y, \end{array} \right. , & \gamma_4 = \left\{ \begin{array}{c} x \longmapsto -1/x, \\ y \longmapsto y/x^{\frac{p+1}{2}}, \end{array} \right. \end{array} \tag{2.2}$$

where $a \in F^\times$ is a primitive $(p-1)^{st}$ root of unity. This group acts transitively on $X(F)$, so it has an orbit of size $|X(F)| = p+1$.

Let $P_1 = (1:0:1)$ and let H be its stabilizer in G. A counting argument shows that H is a solvable group of order $2p(p-1)$ generated by γ_1, $\gamma_2(a)$ and γ_3. By Lemma 1, every point in

$$X(F) = \{(1:0:0), (0:0:1), (1:0:1), ..., (p-1:0:1)\}$$

is ramified over the covering $X \to X/G$ in the sense that each stabilizer $G_P = \mathrm{Stab}_G(P)$ is non-trivial, $P \in X(F)$.

[1]Indeed, say $P \in X(F)$ is not the point at infinity, so $P = (x,y)$, for some $x, y \in F$. By Fermat's Little Theorem, $x^p - x = 0$, so $y = 0$. There are p such points.

It is known (Proposition VI.4.1, [Sti]) that, for each $m \geq 1$, the Riemann-Roch space of $D = mP_1$ has a basis consisting of monomials,

$$x^i y^j, \qquad 0 \leq i \leq p-1,\ j \geq 0,\ 2i + pj \leq m.$$

Lemma 2 *The semisimplification ρ_{ss} of the representation ρ of H acting on $L(D)$ is the direct sum of one-dimensional representations of G.*

Proof: The generator γ_1 acts trivially on the basis of $L(D)$, whereas

$$\gamma_2(a) : \begin{pmatrix} 1 \\ x \\ \vdots \\ x^r y^s \end{pmatrix} \longmapsto \begin{pmatrix} 1 \\ a^2 x \\ \vdots \\ a^{2r+s} x^r y^s \end{pmatrix} = \begin{pmatrix} 1 & 0 & ... & 0 \\ 0 & a^2 & ... & \vdots \\ \vdots & \ddots & & 0 \\ 0 & ... & 0 & a^{2r+s} \end{pmatrix} \begin{pmatrix} 1 \\ x \\ \vdots \\ x^r y^s \end{pmatrix},$$

and

$$\gamma_3 : \begin{pmatrix} 1 \\ x \\ \vdots \\ x^r y^s \end{pmatrix} \longmapsto \begin{pmatrix} 1 \\ x+1 \\ \vdots \\ (x+1)^r y^s \end{pmatrix} = \begin{pmatrix} 1 & 0 & ... & 0 \\ 1 & 1 & ... & \vdots \\ \vdots & \ddots & & 0 \\ 0 & ... & r & 1 \end{pmatrix} \begin{pmatrix} 1 \\ x \\ \vdots \\ x^r y^s \end{pmatrix},$$

where the non-zero terms in the bottom row of the matrix representation of γ_3 are in the last $r+1$ row entries and consist of the binomial coefficients $\frac{r!}{(r-j)!j!}$, $0 \leq j \leq r$. Therefore, the group generated by these matrices is lower-triangular, hence solvable. $\square$

2.1.2 Case $F = GF(p^2)$

Let $F = GF(p^2)$ and let $F_0 = GF(p)$.

The automorphism group $G = \mathrm{Aut}_F(X)$ is a central 2-fold cover of $PGL(2,p)$ and we have a short exact sequence,

$$1 \to Z \to G \xrightarrow{\tau} PGL_2(p) \to 1, \tag{2.3}$$

where Z denotes the subgroup of G generated by the hyperelliptic involution (which coincides with the center of G), by Göb [G]. The group G has order $2|PGL(2,p)| = 2p(p^2-1)$. The following transformations generate G:

$$\gamma_1 = \left\{ \begin{array}{c} x \longmapsto x, \\ y \longmapsto -y, \end{array} \right. , \quad \gamma_2 = \gamma_2(a) = \left\{ \begin{array}{c} x \longmapsto a^2 x, \\ y \longmapsto ay, \end{array} \right.$$
$$\gamma_3 = \left\{ \begin{array}{c} x \longmapsto x+1, \\ y \longmapsto y, \end{array} \right. , \quad \gamma_4 = \left\{ \begin{array}{c} x \longmapsto -1/x, \\ y \longmapsto y/x^{\frac{p+1}{2}}, \end{array} \right.$$

where $a \in F^\times$ is a primitive $2(p-1)^{st}$ root of unity.

Proposition 3 *Let $p > 3$ be a prime.*

(a) Case $p \equiv 3 \pmod 4$:

Let $P_1 = (1:0:1)$ and fix some $P_2 \in X(F) - X(F_0)$. The set of rational points $X(F)$ decomposes into a disjoint union

$$O_1 = X(F_0) = G \cdot P_1, \qquad O_2 = X(F) - X(F_0) = G \cdot P_2,$$

with $|O_1| = p+1$ and $|O_2| = 2p(p-1)$.

(b) Case $p \equiv 1 \pmod 4$:

The automorphism group of X/F acts transitively on $X(F)$ and the stabilizer of any point is a group of order $2p(p-1)$.

Remark 1 *The proof of this proposition is omitted. It has been proven by Bob Guralnick [Gu].*

This and Lemma 1 imply every point in $X(F)$ is ramified for the covering $X \to X/G$.

Let P_1, P_2 be as above and let H_i be the stabilizer of P_i in G (for $i = 1, 2$). We have already seen that H_1 is a solvable group of order $2p(p-1)$ generated by γ_1, $\gamma_2 = \gamma_2(a)$, and γ_3, where $a \in F^\times$ is a primitive $2(p-1)^{st}$ root of unity. As a consequence, $|O_1| = |G \cdot P_1| = |G|/|H_1| = p+1$. In fact, if B denotes the ("Borel") upper-triangular subgroup of $PGL(2,p)$ then $H_1 = \tau^{-1}(B)$. Since B is solvable and any abelian cover of a solvable group is solvable, H_1 is solvable. Since B is not normal in $PGL_2(p)$, H_1 is not normal in G.

Using $H_1 = \langle \gamma_1, \gamma_2, \gamma_3 \rangle$ and the explicit expressions for the γ_i, it can be checked directly that no $g \in H_1$, $g \neq 1$, fixes any $P \in O_2$. Therefore, $H_1 \cap H_2 = \{1\}$. According to the proposition, the stabilizer H_2 of P_2 has order $p+1$. This and $|G| = |H_1| \cdot |H_2|$ implies $G = H_1 \cdot H_2$. In other words, H_2 is a complement of H_1 in G. (As *sets*, $G/H_1 \cong PGL_2(p)/B \cong \mathbb{P}^1(F_0)$.)

By the proposition, the divisor $D_2 = \sum_{P \in O_2} P$ associated to O_2 has degree $2p(p-1) > 2g$, so by the Riemann-Roch theorem, $\dim(L(D_2)) = 2p(p-1) + 1 - \frac{p-1}{2}$.

3 AG code constructions

Throughout this section, we assume X, G, and D are as above and that F is finite.

Let $P_1, ..., P_n \in X(F)$ be distinct points and $E = P_1 + ... + P_n \in Div(X)$ be stabilized by G. This implies that G acts on the support $\mathrm{Supp}(E)$ by permutation. Assume $\mathrm{Supp}(D) \cap \mathrm{Supp}(E) = \emptyset$. Let $C = C(D, E)$ denote the AG code (1.1). The group G acts on C by $g \in G$ sending $c = (f(P_1), ..., f(P_n)) \in C$ to $c' = (f(g^{-1}(P_1)), ..., f(g^{-1}(P_n)))$, where $f \in L(D)$. First, we observe that this map, denoted $\phi(g)$, is well-defined. In other words, if eval_E is not injective and c is also represented by $f' \in L(D)$, so $c = (f'(P_1), ..., f'(P_n)) \in C$, then we can verify $(f(g^{-1}(P_1)), ..., f(g^{-1}(P_n))) = (f'(g^{-1}(P_1)), ..., f'(g^{-1}(P_n)))$. (Indeed, G acts on the set $\mathrm{Supp}(E)$ by permutation.) This map $\phi(g)$ induces a homomorphism of G into the permutation automorphism group of the code $\mathrm{Aut}(C)$, denoted

$$\phi : G \to \mathrm{Aut}(C) \tag{3.1}$$

(Prop. VII.3.3, [Sti]).

The papers [W], [JT], [JK1] investigate ϕ in more detail.

3.1 The kernel of ϕ

To investigate the kernel of this map ϕ, we recall the following notion. Let $H \in Div(X)$ be any divisor. We say that the space $L(H)$ **separates points** if for all points $P, Q \in X$, $f(P) = f(Q)$ (for all $f \in L(H)$) implies $P = Q$ (see [H], chapter II, §7). Recall that if D is ample then $L(D)$ separates points [H].

If G is a group of automorphisms of X defined over F then G induces an automorphism on the image of the evaluation map $\mathrm{eval}_E : L(D) \to F^n$. For this discussion, let us assume this is an injection. (This is not a serious assumption.) To understand the kernel of this map ϕ in (3.1), we'd like to know whether or not $(f(P_1), ..., f(P_n)) = (f(g^{-1}P_1), ..., f(g^{-1}P_n))$ implies

$P_i = g^{-1}P_i$, for $1 \leq i \leq n$. In general, if $L(D)$ separates points then

$$\mathrm{Ker}(\phi) = \{g \in G \mid g(P_i) = P_i,\ 1 \leq i \leq n\}.$$

It is known (see the proof of Prop. VII3.3, [Sti]) that if $n > 2g + 2$ then $\{g \in G \mid g(P_i) = P_i,\ 1 \leq i \leq n\}$ is trivial. Therefore, if $n > 2g + 2$ and D is ample then ϕ is injective.

3.2 Permutation representations

In this subsection, we show how theorems about AG codes can, in some cases, give theorems about representations on Riemann-Roch spaces.

Assume that X/F is a hyperelliptic curve defined over a finite field F of characteristic $p > 2$ with automorphism group $G = \mathrm{Aut}_F(X)$. Let D be a G-equivariant divisor on X, let $\mathcal{O} \subset X(F)$ be a G-orbit disjoint from the support of D, and let $E = \sum_{P \in \mathcal{O}} P$. Let P be the permutation automorphism group of the code $C = C(D, E)$ defined in (1.1).

Theorem 4.6 in [W] (see [JK1] for a more general treatment) implies that if $n = \deg(E)$ and $t = \deg(D)$ satisfy $n > \max(2t, 2g + 2)$ then the map $\phi : G \to P$ is an isomorphism. Using this, we regard C as a G-module. In particular, the (bijective) evaluation map $\mathrm{eval}_E : L(D) \to C$ in (1.2) is G-equivariant. Since G acts (via its isomorphism with P) as a permutation on C, we have proven the following result.

Proposition 4 *Under the conditions above, the representation ρ of G on $L(D)$ is equivalent to a representation ρ' with with property that, for all $g \in G$, $\rho'(g)$ is a permutation matrix.*

The decomposition of $L(D)$ (and hence also C) into indecomposable G-modules, when p divides the order of G, is an interesting open question. For a discussion of a situation related to the case when D is the canonical divisor, see [S].

3.3 Permutation decoding application

If C is a linear code with non-trivial permutation group then this extra symmetry of the code may be useful in decoding. Permutation decoding is discussed, for example, in Huffman and Pless [HP]. It has also been implemented in version 2.0 of the error-correcting codes computer algebra package

GUAVA [GUAVA]. We recall briefly, for the convenience of the reader, the main ideas.

We shall assume that C is in standard form. Let C be a $[n, k, d]$ linear code over F, let $t = [(d-1)/2]$, and let $G = (I_k, A)$ denote the generator matrix in standard form. From this matrix G, it is well-known and easy to show that one can compute an encoder $E : F^k \to F^n$ with image C, and a parity check matrix $H = (B, I_{n-k})$ in standard form, $B = -A^t$.

The key lemma is the following result: Suppose $v = c + e$, where $c \in C$ and $e \in F^n$ is an error vector with Hamming weight $\mathrm{wt}(e) \leq t$. Under the above conditions, the information symbols of v are correct if and only if $\mathrm{wt}(Hv) \leq t$.

Let P denote the permutation automorphism group of C. The permutation decoding algorithm is:

1. For each $p \in P$, compute $\mathrm{wt}(H(pv))$ until one with $\mathrm{wt}(H(pv)) \leq t$ is found (if none is found, the algorithm fails).

2. Extract the information symbols from pv, and use E to compute codeword c_p from them.

3. Return $p^{-1}c_p = \mathrm{Decode}(v)$.

For example, if P acts transitively then permutation decoding will correct at least one error.

The key problem is to find a set of permutations in P which moves the non-zero positions in every possible error vector of weight $\leq t$ out of the information positions. (This set, called a *PD-set*, will be used in step 1 above instead of the entire set P.)

Example 5 *We give an example using [GUAVA] of an AG code for which permutation decoding seems to apply.*

This is an example of a $[7, 3, 5]$ *one-point AG code over* $GF(7)$ *arising from the hyperelliptic curve* $y^2 = x^7 - x$. *The following GUAVA (version 2.0) commands help investigate this example.*

```
p:=7;
F:=GF(p);
R := PolynomialRing(F,2);
indets := IndeterminatesOfPolynomialRing(R);x:=indets[1]; y:=indets[2];
```

```
Pts:=AffinePointsOnCurve(y^2-x^p+x,R,F);; Length(P);
C:=OnePointAGCode(y^2-x^p+x,Pts,4,R);
MinimumDistance(C);
G:=GeneratorMat(C);
PutStandardForm(G);
Display(G);
P:=PermutationAutomorphismGroup(C);
IdGroup(P);
Wt2:=ConstantWeightSubcode(WholeSpaceCode(p,GF(p)),2);
for c in C do
 e:=Random(Wt2);; v:=PermutationDecode(C,c+e);
 if c<>v then Print("\n Decoding fails. c,v = ",c," ",v,"\n"); if;
od;
```

After many runs, no decoding failures were found, in support for Conjecture 6 below. This code has generator matrix in standard form given by

$$G = \begin{pmatrix} 1 & 0 & 0 & 6 & 4 & 6 & 4 \\ 0 & 1 & 0 & 6 & 3 & 3 & 1 \\ 0 & 0 & 1 & 3 & 1 & 6 & 3 \end{pmatrix}.$$

Moreover, the permutation automorphism group of the code is a group of order 42 *generated by* $(1,5,4)(2,3,7)$ *and* $(1,3,2,6,7,4)$. *The elements of* $P - \{1\}$ *can be used as a PD-set.*

The reader may experiment him/herself by using similar GUAVA commands to compute other examples[2].

Define a *codeword operation* to be either

- a syndrome computation (i.e., Hv, where H is a parity check matrix and v is a received word),
- a vector comparison (i.e., evaluating a Boolean $v = v'$, where v, v' are two vectors),

[2]The GUAVA command `PermutationAutomorphismGroup(C)`, written in GAP's own interpretive language, is probably too slow for $p > 11$, as the case $p = 13$ took over 1 day on an AMD64 bit linux machine. The command `ConstantWeightSubcode(WholeSpaceCode(p,GF(p)),2)` is much faster on a suitably configured linux/unix machine since in that case GUAVA can call on relatively fast binary programs written in C.

- a permutation of coordinates (i.e., pv, where $p \in S_n$ is an element of the symmetric group on n letters and $v \in F^n$),
- a Hamming weight computation (i.e., $\mathrm{wt}(v)$).

Conjecture 6 *For one-point AG codes C in standard form associated to $y^2 = x^p - x$ over $GF(p)$ of length $n = p$, permutation decoding always applies. Its complexity in codeword operations is at worst the size of the permutation group of C, which is $O(p^2) = O(g^2) = O(n^2)$.*

Note that the size of the alphabet is changing as $n \to \infty$. This conjectured complexity is to be compared with the decoding complexity of $O(n^3)$ field operations for the well-known interpolation algorithm (see for example §3.3.1 of [TV] or [Sti] or [HvLP]).

Conjecture 7 *Assume $p \equiv 3 \pmod 4$, $p > 3$. For one-point AG codes $C = C(m \cdot (1:0:0), O_2)$ of length $n = 2p(p-1)$ in standard form associated to the hyperelliptic curve X over $GF(p^2)$ defined by $y^2 = x^p - x$, permutation decoding always applies. We conjecture that, if the points in $X(F)$ are arranged suitably then the image of $\mathrm{Stab}_G((1:0:0)) \subset G = \mathrm{Aut}_F(X)$ in the permutation group of C may be used as a PD-set. Its complexity in codeword operations is at worst $O(p^2) = O(g^2) = O(n)$.*

If true[3], this seems to beat the complexity of other decoding algorithms, such as those surveyed in [HvLP]. (When comparing estimates, note other complexity bounds of decoding algorithms usually assume the size of the ground field q is fixed. Here it is a variable.)

Finally, we observe that known methods for computing the automorphism group of a linear code in general are not very efficient (see for example Leon [L]). However, the equations (2.2) allow one to compute in this case an action of G on the code efficiently.

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